

GHANA COMMUNICATION TECHNOLOGY UNIVERSITY



FACULTY OF COMPUTING and INFORMATION SYSTEMS BSc. COMPUTER SCIENCE & SOFTWARE ENGINEERING(Level 100) END OF SEMESTER EXAMINATIONS, JULY 2022

MATH 105: LINEAR ALGEBRA

Duration: 2.5HRS

Instructions: *The exam is made up of two sections (Section A and section B). Students are required to attempt all questions in section A and any two in section B.*
All answers must be in the answer booklet provided

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SECTION A
Attempt ALL questions
[30 marks]

1. The solution of the given matrix equation $A(x, y) = \begin{bmatrix} 3 & 0 & 2 \\ 6 & 1 & 1 \\ 2 & 8 & 91 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is
- (a) $x = 1, y = 1, z = 2$
 - (b) $x = 0, y = 0, z = 0$
 - (c) $x = 3, y = -1, z = -1$
 - (d) $x = 0, y = -2, z = 4$
 - (e) none of the above
2. Which of the following is true for matrices?
- (a) $(AB)^{-1} = B^{-1}A^{-1}$
 - (b) $(A^T) = A$
 - (c) $AB = BA$
 - (d) $A \times I = I$
 - (e) none of the above
3. Let $T : V \rightarrow V$ be a linear operator and $T(x) = \lambda x$ for some scalar λ . Then x is called
- (a) an eigenvalue of T .
 - (b) an eigenvector of T .
 - (c) an eigenspace of T .
 - (d) spectrum
 - (e) none of the above
4. The determinant of the matrix whose eigenvalues are 7, 1, 9 is given by
- (a) 7
 - (b) 63
 - (c) 9
 - (d) 17
 - (e) none of the above
5. The sum of two symmetric matrices is also a symmetric matrix.

- (a) True
- (b) False
- (c) none of the above

6. Find the values of x and y in the matrix below if the matrix is a skew symmetric matrix. $A(x, y) =$

$$\begin{bmatrix} 0 & x+y & 6 \\ 3 & 0 & 9 \\ x & 9 & 0 \end{bmatrix}$$

- (a) $x = -6, y = 3$
- (b) $x = 3, y = 3$
- (c) $x = 6, y = -3$
- (d) $x = 0, y = 3$
- (e) none of the above

7. Let A be a 5×5 matrix with real entries and $x \neq 0$. Then, the vectors $x, Ax, A^2x, A^3x, A^4x, A^5x$ are

- (a) linearly independent
- (b) linearly dependent
- (c) linearly independent if and only if A is symmetric
- (d) linear dependence/independence cannot be determined from given data
- (e) none of the above

8. Let A, B, C, D be square matrices of the same size. Then, $\begin{vmatrix} A & B \\ C & D \end{vmatrix} = \begin{vmatrix} D & C \\ B & A \end{vmatrix}$

- (a) True
- (b) False
- (c) none of the above

9. Suppose the determinant of a 2×2 matrix A is equal to 5. What is the determinant of $2A$?

- (a) 5
- (b) 10
- (c) 20
- (d) 25
- (e) There is insufficient information to answer the question.

10. Find the value of x, y, z, w in the matrix equation $5 \begin{bmatrix} x & z \\ y & w \end{bmatrix} - \begin{bmatrix} z & 5 \\ 7 & w \end{bmatrix} = \begin{bmatrix} 2 & 10 \\ 3 & 2x+y \end{bmatrix}$

- (a) $x = 1, y = 2, z = 3, w = 1$
- (b) $x = -1, y = 2, z = -3, w = 1$
- (c) $x = 1, y = -2, z = 3, w = -1$
- (d) $x = 2, y = 1, z = 1, w = 3$
- (e) none of the above

11. The determinant of $\begin{bmatrix} a-x & b-y & c-z \\ x & y & z \\ a & b & c \end{bmatrix}$ is

- (a) 0
- (b) $a - x$
- (c) $b - y$
- (d) $c - z$
- (e) none of the above

12. Which of the following is an eigenvector of $A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix}$

- (a) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$
- (b) $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$
- (c) $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$
- (d) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- (e) none of the above

13. Consider the matrix $A = \begin{bmatrix} 2 & 3 \\ x & y \end{bmatrix}$. If the eigenvalues of A are 4 and 8, then

- (a) $x = 4, y = 10$
- (b) $x = 5, y = 8$
- (c) $x = -3, y = 9$
- (d) $x = -4, y = 10$
- (e) none of the above

14. If A is an $n \times n$ matrix that is not invertible, then the linear system $Ax = 0$ has

- (a) infinitely many solutions
 - (b) unique solution
 - (c) no solution
 - (d) finitely many solutions
 - (e) none of the above
15. Let A be an $n \times n$ matrix. The linear system $Ax = 4x$ has a unique solution if and only if . . . is an invertible matrix.
- (a) A
 - (b) $A - 4I$
 - (c) $A + 4I$
 - (d) $4A$.
 - (e) none of the above
16. An $n \times n$ matrix has
- (a) at most n distinct eigenvalues
 - (b) at least n distinct eigenvalues
 - (c) exactly n distinct eigenvalues
 - (d) exactly $n+1$ distinct eigenvalues
 - (e) none of the above
17. If the size of A is 4×3 , B is 4×5 and C is 7×3 , what is the size of $(A^T B)^T C^T$
- (a) 5×3
 - (b) 4×5
 - (c) 5×7
 - (d) 4×3
 - (e) not defined
18. A matrix B has a dimension of 2×2 . If the eigenvalues of the matrix are 4 and 7, what would be the eigenvalues of B^2
- (a) 11 and 28
 - (b) 16 and 49
 - (c) 4 and 7
 - (d) 4 and 14

- (e) none of the above
19. If $\begin{bmatrix} 2 & 3 & 4 \\ x-3 & -2 & -1 \\ x-2 & -1 & -5 \end{bmatrix}$ is symmetric, then $x =$
- (a) 3
(b) 6
(c) 8
(d) 0
(e) none of the above
20. Are the vectors $v = (2, 0, -1)$, $w = (4, 0, 7)$, and $s = (-1, 1, 4)$ linearly independent in R^3 ?
- (a) linearly dependent
(b) linearly independent
(c) Data not complete
(d) linearly independent in R^2
(e) none of the above
21. A matrix B is said to be similar to matrix A , if $B = P^{-1}AP$.
- (a) True
(b) False
(c) none of the above
22. If $T_A : R^n \rightarrow R^n$ and if $T_A(x) = 0$ for every vector $x \in R^n$, then A is
- (a) the $n \times n$ zero matrix
(b) the $n \times n$ identity matrix
(c) an elementary matrix
(d) cannot be determined
(e) none of the above
23. Solve for x and y in the matrix equation below: $\begin{bmatrix} 3 & -4 & y \\ 1 & 2 & -1 \end{bmatrix} + \begin{bmatrix} x-y & 4 & 2 \\ 4 & x & -8 \end{bmatrix} = \begin{bmatrix} 10 & 3 & -1 \\ 2 & 6x & -9 \end{bmatrix} - \begin{bmatrix} 4 & 3 & -1 \\ -3 & 3x & 0 \end{bmatrix}$
- (a) $x = -1, y = -2$
(b) $x = -1, y = 2$

(c) $x = 1, y = -2$

(d) $x = 1, y = 2$

(e) none of the above

24. Matrix multiplication is commutative

(a) True

(b) False

(c) none of the above

25. Let A and B be the 2×2 matrices. If $A = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ Which of the following statements is most accurate?

(a) Neither A nor B is diagonalizable.

(b) Both A and B are diagonalizable.

(c) A is diagonalizable but B is not.

(d) B is diagonalizable but A is not.

(e) none of the above

26. For all square matrices A and B , is it true that $\det(A + B) = \det(A) + \det(B)$

(a) True

(b) False

27. Find the eigenvalues of the matrix $\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$

(a) 4, 1

(b) -1, -4

(c) 3, 5

(d) -3, -5

(e) none of the above

28. A matrix A is diagonalizable if and only if it has linearly independent eigenvectors.

(a) True

(b) False

(c) none of the above

29. If A is $m \times n$ matrix such that AB and BA are both defined, then B is a matrix of dimension

- (a) $m \times n$
- (b) $n \times n$
- (c) $n \times m$
- (d) $m \times m$
- (e) none of the above

30. For what value of k do the system of linear equations $2x + 3y = 1$ and $4x + 6y = k$ have infinitely many solutions

- (a) $k = 0$
- (b) $k = 1$
- (c) $k = 2$
- (d) $k \neq 2$
- (e) none of the above

SECTION B

Attempt ANY TWO Questions

QUESTION 1

1a Find AB and express the second column of AB as a linear combination of the columns of A , given

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 2 & 0 \end{bmatrix}, B = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 5 & 1 \end{bmatrix}$$

[5 marks]

1b Let

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 3 & 0 \\ 1 & 1 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ 2 & 0 \\ 1 & -2 \end{bmatrix}, C = \begin{bmatrix} 2 & -1 \\ 1 & 1 \\ 0 & 3 \end{bmatrix}$$

- i Find $|A|$
- ii Find a matrix X such that $AX + B = C$.
- iii Is it possible to find a matrix Y such that $YA + B = C$? Explain.

[5 marks]

1c Under which conditions does the system

$$\begin{aligned} x - y + (a + 3)z &= a^3 - a - 7 \\ -x + ay - az &= a \\ 2(a - 1)y + (a^2 + 2)z &= 8a - 14 \end{aligned}$$

- i has a unique solution
- ii No solution
- iii Infinitely many solutions

1d Prove that similar matrices have the same eigenvalues.

[5 marks]

Question 2

2a A system of Linear equations is given as

$$\begin{aligned} 2x + z &= 1 \\ y - 2z &= 0 \\ x + y + z &= -1 \end{aligned}$$

- i. Put the system in the matrix equation form $AX = B$ [1 mark]
- ii. Find the RREF of the system [2 marks]
- iii. Write the general solution to the system [2 marks]
- iv. Write the trivial solution to the corresponding homogeneous system of equations [2 marks]

2b Let u and v be two solutions of the non-homogenous system $Ax = B$. Show that $u - v$ is a solution to the homogenous system $Ax = O$. [2 marks]

2c Let $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 0 \end{bmatrix}$

- i. Find all the spectrum of A [1 mark]
- ii. Find all the eigenspace of A [2 marks]
- iii. Is A diagonalizable?
- iv. If possible, find a nonsingular matrix P and a diagonal matrix D such that $P^{-1}AP = D$. [2 marks]

[2 marks]

Question 3

3a Suppose that $S = v_1, v_2, v_3$ is a linearly independent set of vectors in a vector space V . Show that $T = w_1, w_2, w_3$ is also linearly independent set, where $w_1 = v_1 + v_2 + v_3$, $w_2 = v_2 + v_3$, and $w_3 = v_3$. [3 marks]

3b Find the dimension of the subspace of all vectors of the form $(a + c, a + b + 2c, a + c, a - b)$ where $a, b, c \in \mathbb{R}$. [4 marks]

3c Prove that if u, v is a linearly dependent set, then u is perpendicular to v . [4 marks]

3d Let $A = \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & -1 & 0 & 0 & 2 \\ 2 & 1 & 6 & 0 & 1 \end{bmatrix}$

- i Find rank of A,
- ii. Find the nullity of A.

[4 marks]

[30 marks]